erative process is likely to degenerate over time (*"clueless generator"*). Using fixed-point analysis across iterations, they find that when the proportion of clean data in the mix is sufficiently high, the generator, under certain technical conditions, retains the capability to learn. A recent paper (Fan et al., 2023) empirically observe deteriorated scaling laws when training on synthetic data for text-to-image models. [3]

To our knowledge, our work is the first to theoretically and empirically analyze model collapse in the context of scaling laws and emergent abilities to provide a rich new landscape of AI-data induced phenomena.

2. A Deterministic Infinite Memory Model

Here, we present the core of our theory for the simplest case of (i) *infinite memory* and (ii) *a deterministic ground truth* labeling function $i \mapsto y_i$, studied by Hutter (2021) (the *"Hutter LLM"*). Both restrictions will be lifted in later sections, where we also analyze an *probabilistic autoregressive* version (Section 4) and *limited memory* models (Section 5). Token i is drawn according to the Zipf law in Equation (1), which e.g. models distribution of various metrics in language. Another interpretation of the appearance of a power-law is offered by the "Quantization Hypothesis" paper of Michaud et al. (2023): one may think of each i as some discrete skill, needed to solve a problem for example; thus, the skills occur at different rates p_i. The shape parameter $\beta > 1$ controls the length of the tail of this distribution: bigger values of β correspond to longer tails.

2.1. What Causes Model Collapse ?

Tail Cutting. As mentioned, deliberate choices in the AI generation algorithm (like top-p^{inf} or top-k^{inf} next token prediction) immediately lead to a chopped tail at k. When viewed as skills, we can say that only the kth most frequent outcomes ("skills") are considered. But even when no tails are cut deliberately, the finite size T_0 of the training set (sampling bias) induces an effective tail-cutting. This can be seen as follows: Sample an iid dataset of size T_0, and estimate the histogram p_{AI}; this new distribution plays the role of an AI data-generator. An integer i appears in the support of p_{AI} a number of times which is $T_0 p_i$ on average. Roughly speaking[4], this means that the support of p_{AI} is $\{i \mid p_i \geq C/T_0\} = \{i \mid i \leq k\}$, where

$$k = k(T_0) \asymp T_0^{1/\beta}. \tag{6}$$

Therefore, the transformation $p \to p_{\text{AI}}$ amounts to chopping off the tail of p at rank k, where k is as given above.

[3] A more detailed description of related and prior work can be found in Appendix A

[4] This can be made rigorous via standard concentration arguments.

Tail Narrowing. Figure 2 (for Llama2) shows that in addition to tail cutting, tail narrowing effects happen during AI-generation. One mechanism for this is lowered temperature during next-token prediction. Assume a softmax distribution on the logits z_i for the ith token: $p_i = e^{z_i}/\sum_j e^{z_j}$. Define $q_i^\tau = e^{z_i/\tau}/\sum_j e^{z_j/\tau}$ for general temperature τ. Then $p_i \asymp i^{-\beta}$ morphs into $q_i^\tau \asymp i^{-\beta/\tau}$ (to first order). We see that temperature scaling directly causes narrowing of tail for $\tau > 1$. Other mechanisms can come to play: for instance, for autoregressive models with perplexity, token-wise tail cutting can result in tail narrowing for sequence-perplexity (see Figure 35 and discussion in Appendix I).

2.2. A New Scaling Law in the Hutter LLM

For a deterministic ground-truth labelling function $i \mapsto j_i$, consider a downstream Hutter "LLM" (Hutter, 2021)

$$\widehat{f}(i) := \begin{cases} j_i, & \text{if } (i, j_i) \in \mathcal{D}_T, \\ \perp, & \text{otherwise,} \end{cases} \tag{7}$$

constructed on a sample $\mathcal{D}_T := \{(i_t, j_{i_t}) \mid t \in [T]\}$ of size T from unmitigated Zipf distribution p (1), the test error obeys the following scaling law Hutter (2021)

$$E_{test} \asymp T^{-(1-1/\beta)}. \tag{8}$$

Now, let q be a k-tail-cutting version of p, i.e $q_i \propto p_i$ if $i \leq k$ and $q_i = 0$ otherwise. When constructed ("trained") on $\mathcal{D}_T$ of size T, now from q, the test error (w.r.t to the true data distribution p) of this model is

$$E_{test} := \mathbb{P}_{i\sim p}(\widehat{f}(i) \neq j_i) = \sum_{i\geq 1} p_i \mathbb{P}(\widehat{f}(i) \neq j_i). \tag{9}$$

That is, we train on data from the AI distribution q and test on original distribution p. We prove the following scaling law for tail cutting (all proofs are relegated to Appendix C):

Theorem 2.1. *Consider long-tail real-world data with exponent $\beta > 1$, and let the cutoff for AI-generated data be k. Then, for large k and T samples from the AI, the test error of the downstream "LLM" scales like so* $E_{test} \asymp T^{-(\beta-1)/\beta} + k^{-(\beta-1)} \asymp \min(T, k^\beta)^{-(\beta-1)/\beta}$.

Thus, as soon as $T \gtrsim k^\beta$, the AI-generated sample size T ceases to be a "scalable" resource: collecting more AI-generated samples will not improve the performance of the downstream model, i.e performance plateaus and we lose scaling. The result is illustrated empirically in Figure 3, left and Figure 8 (Appendix B).

When we assume that the AI-generator itself was trained on T_0 samples, we get a similar loss of scaling stemming from the tail cutting from finite sampling bias (Equation (6)):

Corollary 2.2 ("Finite Initial Sample Size")**.** *With $c = 1 - 1/\beta$, it holds that*

$$E_{test} \asymp T^{-c} + T_0^{-c}. \tag{10}$$

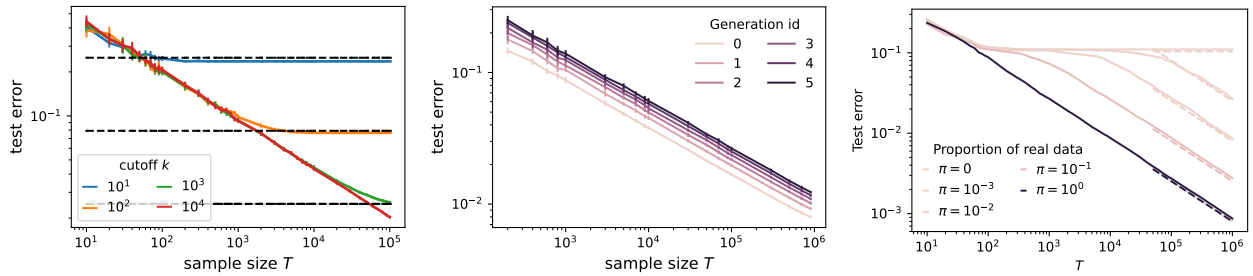

Figure 3. **Illustration of Our Main Results for Simplified LLMs. Left plot.** Empirical confirmation of the double scaling law. The true distribution of the data is Zipf with exponent $\beta = 3/2$. Broken lines correspond to $k^{-(\beta-1)}$, for varying T and different values of k. **Middle plot.** Model collapse over multiple generations. Again $\beta = 3/2$, $T_0 = T$ across all generations with no additional tail-cutting, regeneration for 5 times. **Right plot.** Notice the grokking behavior, as perfectly predicted by the Theorem 3.2. For any given value π for the proportion of real data, the broken lines are asymptotes $E_{test} \asymp (\pi T)^{-c}$ and each plateau has length of order k^{β}/π, both predicted by the theorem. See Figure 10 for similar results with other values of k.

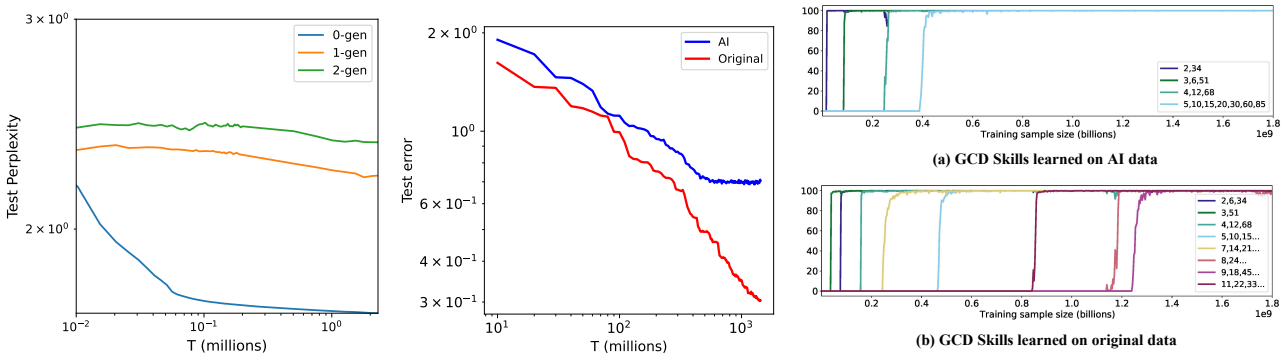

Figure 4. **Experimental Results (Details in Section 6). Left plot.** The scaling law for finetuning `Llama2-7B` on the Wikitext-103 dataset. '0-gen' utilizes the original data, while subsequent generations use data generated by the previous model. **Middle plot.** Scaling law of the transformer model trained to predict GCD of two integers. Data is synthesized from a 0th generation model trained on $300K$ samples. Note the tapered-off scaling of the model trained on synthesized data, as predicted by our theory. **Right plot.** "Skills" (bursts of new GCDs) learned by the GCD-transformer on original (bottom) and AI data (top). We see how the disappearance of scaling leads to the disappearance of abilities, mastered by the model trained on clean data.

These theoretical are empirically confirmed in the Figure 9. In the case of tail narrowing, the scaling behavior changes; instead of a plateau, we obtain a slower decay rate:

Corollary 2.3 ("Tail Narrowing"). *In the setting of Theorem 2.1, consider AI-generated data to also be long-tail data, albeit with smaller exponent $\beta' \in (1, \beta)$. Then, the downstream Hutter LLM trained on AI-generated data will scale as $E_{test} \asymp T^{-(\beta-1)/\beta'}$.*

2.3. Collapse Over Multiple Generations of AI Data

We now examine the cumulative impact of prior loss of scaling across multiple generations. Consider n-fold recursive AI data-generation, i.e

$$p \to p_{\mathrm{AI}(1)} \to p_{\mathrm{AI}(2)} \to \ldots \to p_{\mathrm{AI}(n)}. \tag{11}$$

Each arrow corresponds to drawing a sample of size T_0. If we iterate n times the argument leading to (10), we get the following scaling for the test error $E_{test}^{(n)} = E_{test}^{(n)}(T)$ for learning on T samples from the nth generation and testing on the true data distribution,

$$\begin{aligned} E_{test}^{(n)} &\asymp T^{-c} + \underbrace{T_0^{-c} + \ldots + T_0^{-c}}_{n \text{ times}} \\ &= T^{-c} + nT_0^{-c} = T^{-c}\left(n(T/T_0)^c + 1\right), \end{aligned} \tag{12}$$

where $c := 1 - 1/\beta$. Only in the context of model collapse across multiple generations, T is the size of the AI data in the final step and T_0 is the data a model trained in all preceding steps. We deduce the following result.

Theorem 2.4 (Informal). *Model collapse (as spoken of in the literature) occurs iff $n \gg (T_0/T)^c$.*

For example, if $T_0 \gg T$ (e.g $T_0 \geq CT \log T$) and n is constant (e.g $n = 25$), then model collapse will not occur if we learn on the nth generation of AI data. On the other hand, if $T_0 \lesssim T$, then model collapse will eventually occur.

In particular, taking $T_0 \asymp T$, we get

$$E_{test}^{(n)} \asymp C_n T^{-c} \asymp nT^{-c}. \tag{13}$$

Note how the loss scales linearly with the number of generations. Figure 3, middle, illustrates how an increased number

of generations moves the loss scaling curve progressively to the right. This leads to eventual model collapse.

3. Mitigating Model Collapse via Data Mixing

Here we explore the possibility of alleviating model collapse via the acquisition of even a tiny amount of data from the true data distribution, to complement AI polluted data. We study two phenomena: (1) In the case of mixing π-fraction of the original data with a $(1-\pi)$ fraction of AI-generated data we exhibit a startling "grokking" phenomenon where test loss plateaus with increasing training data to finally decrease again according to the scaling law of the original model, and (2) in the scenario where we would like to compensate for missing "tail", we acquire some data from the tail of the original distribution to show that this needs to be done with caution: getting data from "too deep" in the tail is worthless while data closer to the precise "missing" tail can be beneficial. All proofs can be found in Appendix C.

3.1. Acquiring Missing Tail

To counter the effect of tail cutting and the resulting plateau in scaling, we might resort to adding curated data that would emphasize the tail. The following Theorem 3.1 studies this effect; it shows, in particular that if we "overshoot" and only curate tail that is too deep, our efforts will be worthless. Rather, there is a fine line around the chopped tail k (within a factor of $(1+o(1))$ of k), where we need to place our data curation efforts to get the scaling back.

Suppose we "buy" a chunk of the tail of the real data distribution corresponding to $i = N, N+1, \ldots$; let the distribution be π (thus, supported on $\{N, N+1, \ldots\}$). Now, let k, N, and T tend to infinity such that $N/k \to C$, with $C \in [1, \infty]$. We have the following sharp phase-transition.

Theorem 3.1. *(A) If $C = 1$, e.g if $N = k + \sqrt{k}$, then $E_{test} \asymp T^{-c}$. That is, we perfectly anneal the tail-chopping effect of AI-generated data.*

(B) If $C > 1$, then $E_{test} \asymp T^{-c} + k^{-\alpha}$ (which recovers the result of Theorem 2.1), and so "buying" the Nth tail of the real data distribution is worthless.

3.2. A Grokking Phenomenon

Here we show how even small amounts of original data can mitigate the above "scaling law collapse" by introducing a grokking phenomenon where test error plateaus and eventually continues to decline.

Theorem 3.2 (Grokking with Tail Cutting)**.** *Consider a sample of size T of which a proportion π comes from the true distribution p and the remainder comes from a version p' of p with its tail chopped off at rank k. We have the following scaling laws for the Hutter LLM define din* (7).

(A) ***Early-Stage Dynamics.*** *For $T \ll k^{\beta}/\pi$, it holds that*

$$E_{test} \asymp T^{-(1-1/\beta)} + k^{-(\beta-1)}. \tag{14}$$

Thus, during this stage, the money spent on acquiring some clean data is not amortized!

(B) ***Later-Stage Dynamics.*** *As soon as $T \geq Ck^{\beta}/\pi$ (where C is an absolute constant), it holds that*

$$E_{test} \asymp (\pi T)^{-(1-1/\beta)}. \tag{15}$$

Thus, during this stage, we recover the unpolluted sample-size law scaling $T^{-(1-1/\beta)}$, up to within a multiplicative constant $\pi^{-(1-1/\beta)}$ (which can be seen as an increase in the price of data). For fixed T and tunable π, this error rate scales like $\pi^{-(1-1/\beta)}$, which is yet another scaling law.

Effectively, the above theorem predicts that for any fixed $\pi \in (0,1)$ –no matter how small– the test error grokks w.r.t sample size T. The result is empirically confirmed in Figure 3, right (see Figure 10 for another illustration).

We experimentally confirm this new phenomenon for transformer models trained to calculate the GCD (see Appendix G), which indicates its applicability for a wider class of LLMs with underlying deterministic ground truth, like for arithmetic tasks.

In Appendix C.4 we state and prove a similar theorem in the case of *tail narrowing* of synthetic data.

Benefits of Mixing with AI Data. The above machinery allows us to analyze a particular regime where AI-data can help improve performance.

Taking $T = T_{real} + T_{AI}$ and $\pi = T_{real}/T$, we have the following important corollary of Theorem 3.2.

Corollary 3.3. *For $T_{real} \ll k^{\beta}$, it holds that $E_{test} \asymp (T_{real} + T_{AI})^{-(1-1/\beta)} + k^{-(\beta-1)}$.*

Figure 5 illustrates how AI data can boost performance, up to a certain point, when its benefits plateau. This result might contribute to our understanding of why, sometimes, adding AI-generated data might lead to better models, especially when generated by a stronger model (e.g. He et al. (2023); Shipard et al. (2023); Bansal & Grover (2023); Lin et al. (2023)). See Appendix A for more references.

4. A Tailed Bigram Model

We will now proceed to a more complex model, bringing us closer to capturing the *probabilistic* and *autoregressive* nature of LLMs (next token prediction). In this Section we will define the data generating process, define the new model (Hutter++), and establish that the original scaling law (with clean data) still holds. We then proceed to show similar loss of scaling for AI-data.